# Current, Resistance, and Electromotive Force

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# Learning Outcomes

**In this chapter, you'll learn…**

- the meaning of electric current, and how charges move in a conductor.
- how to calculate the resistance of a conductor from its dimensions and its resistivity or conductivity.
- how an electromotive force (emf) makes it possible for current to flow in a circuit.
- how to do calculations involving energy and power in circuits.
- how to use a simple model to understand the flow of current in metals.

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# Introduction

- Electric circuits contain charges in motion.
- In a flashlight, the amount of current that flows out of the bulb is the same as the amount that flows into the bulb.
- It is the **energy** of the charges that decreases as the current flows through light bulbs.
- Circuits are at the heart of modern devices such as computers, televisions, and industrial power systems.

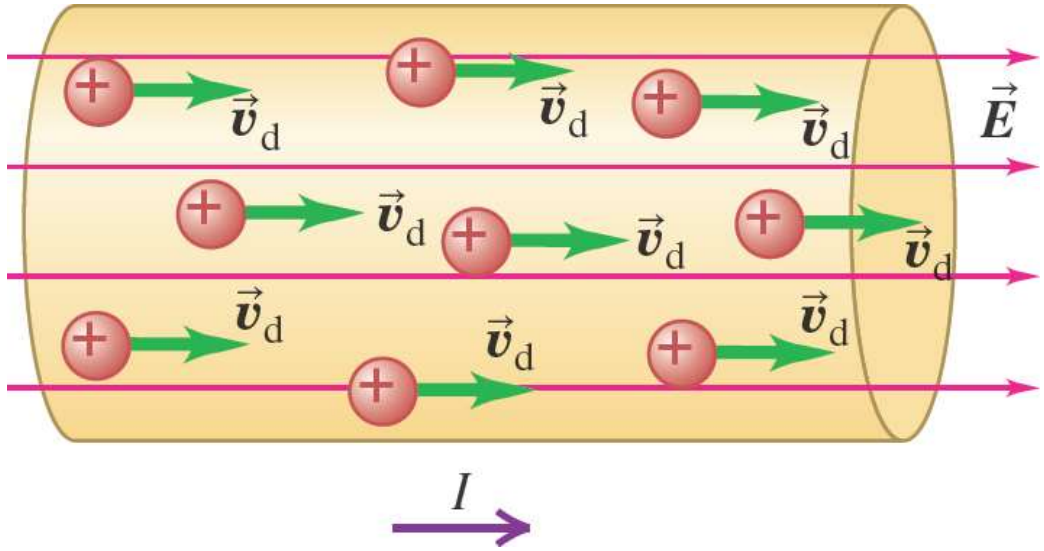


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# Current

- A **current** is any motion of charge from one region to another.




Rate at which charge flows through area

**Current** through an area

$$I = \frac{dQ}{dt} = n|q|v_d A$$

Drift speed

Cross-sectional area

Charge per particle

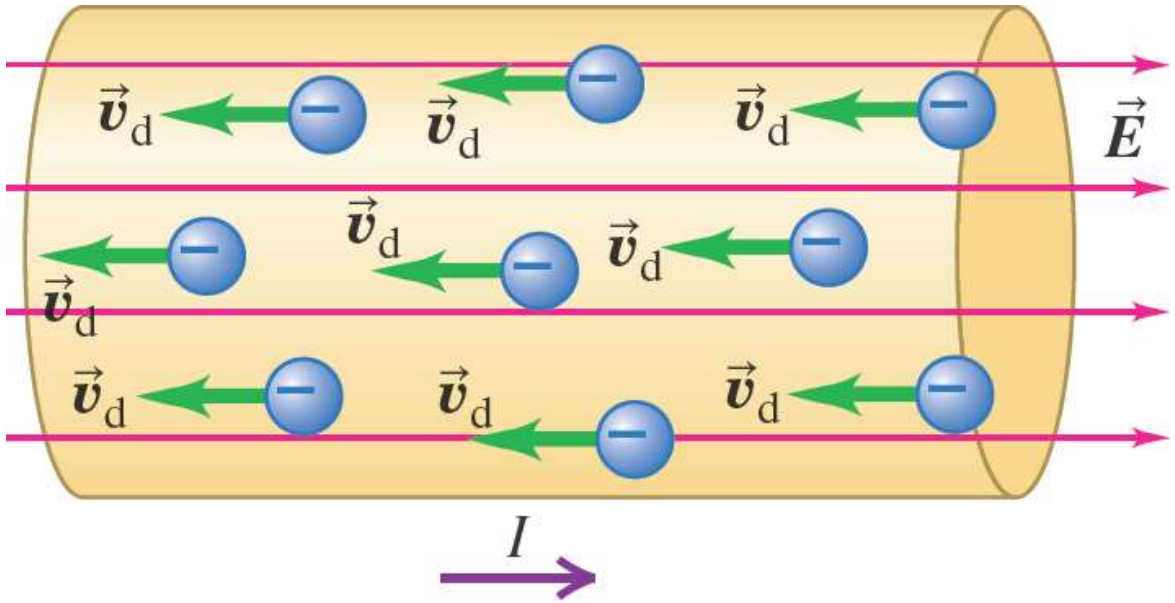
Concentration of moving charged particles

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# Direction of Current Flow

- A current can be produced by positive or negative charge flow.
- **Conventional current** is treated as a flow of positive charges.
- In a metallic conductor, the moving charges are electrons—but the **current** still points in the direction positive charges would flow.




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# Signs of Charge Carriers

- In general, a conductor may contain several different kinds of moving charged particles.
- An example is current flow in an ionic solution.
- In the sodium chloride solution shown, current can be carried by both positive sodium ions and negative chlorine ions
- The total current *I* is found by adding up the currents due to each kind of charged particle.

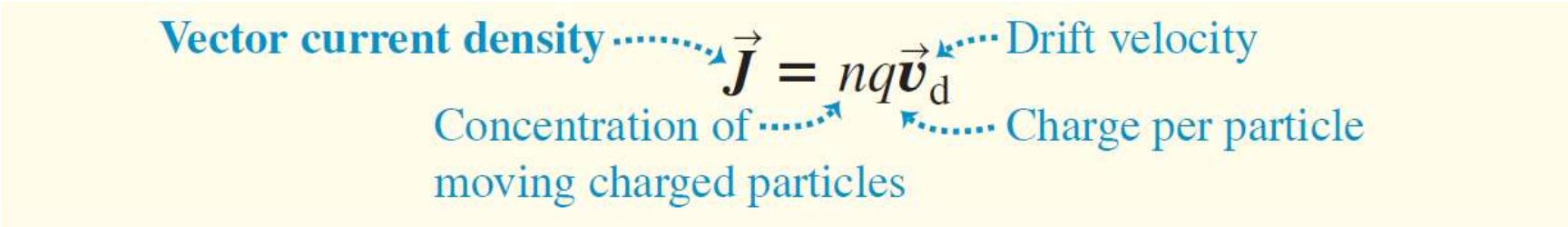


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# Current Density

- We can define a **vector** current density that includes the direction of the drift velocity:



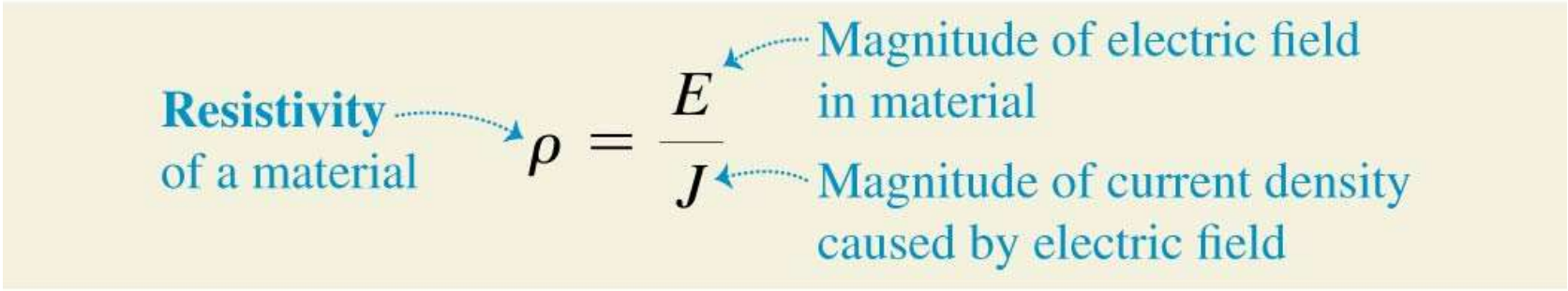

- The vector current density is always in the same direction as the electric field, no matter what the signs of the charge carriers are.

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# Resistivity

- The **resistivity** of a material is the ratio of the electric field in the material to the current density it causes:




- The **conductivity** is the reciprocal of the resistivity.
- The next slide shows the resistivity of various types of materials.

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# Resistivities at Room Temperature (20 degrees Celsius)

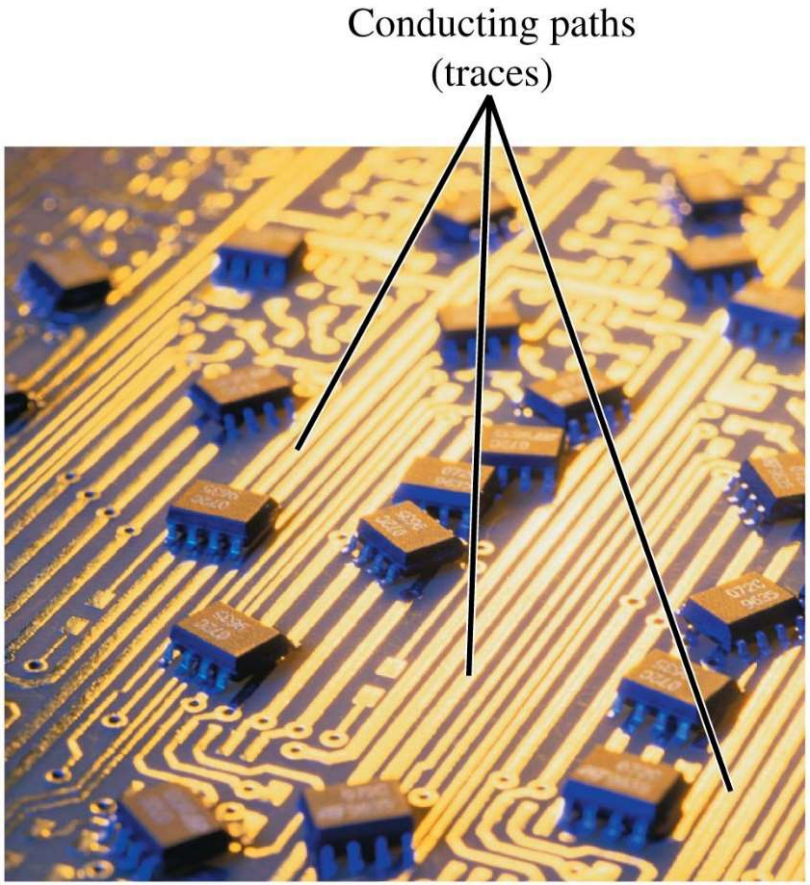
| | Substance | $\rho$ ($\Omega \cdot$ m) |
|---|---|---|
| Conductors | Copper | $1.72 \times 10^{-8}$ |
| | Gold | $2.44 \times 10^{-8}$ |
| | Lead | $22 \times 10^{-8}$ |
| Semiconductor: | Pure carbon (graphite) | $3.5 \times 10^{-5}$ |
| Insulators | Glass | $10^{10} - 10^{14}$ |
| | Teflon | $>10^{13}$ |
| | Wood | $10^{8} - 10^{11}$ |

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# Circuit Boards and Resistivity

- The copper “wires,” or traces, on this circuit board are printed directly onto the surface of the dark-colored insulating board.
- Even though the traces are very close to each other, the board has such a high resistivity that essentially no current can flow between the traces.

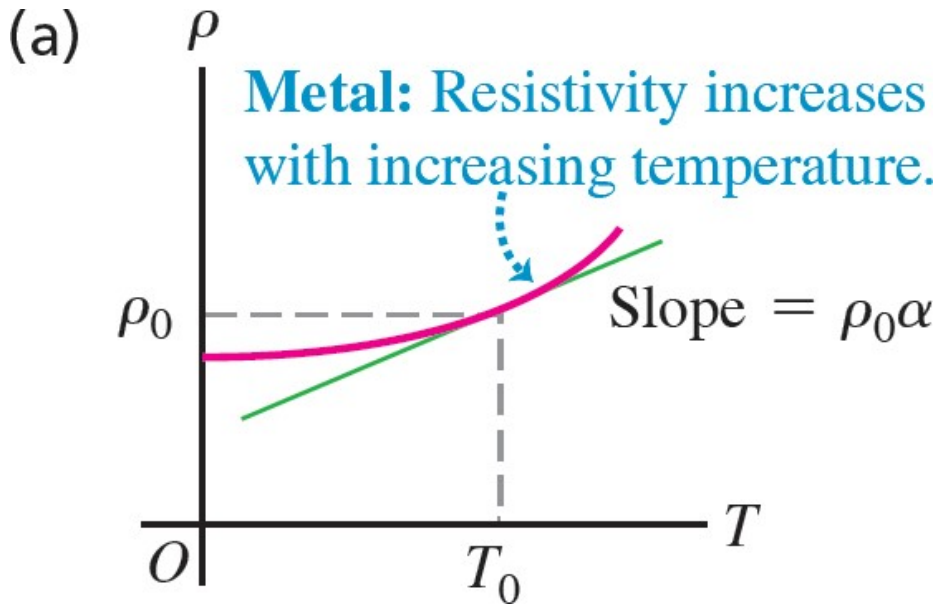



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# Resistivity and Temperature (1 of 2)

- The resistivity of a **metallic** conductor nearly always increases with increasing temperature.
- Over a small temperature range, the resistivity of a metal can be represented approximately:




**Temperature dependence of resistivity:**

$$\rho(T) = \rho_0[1 + \alpha(T - T_0)]$$

$\rho(T)$: Resistivity at temperature $T$

$\alpha$: Temperature coefficient of resistivity

$\rho_0$: Resistivity at reference temperature $T_0$

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# Temperature Coefficients of Resistivity

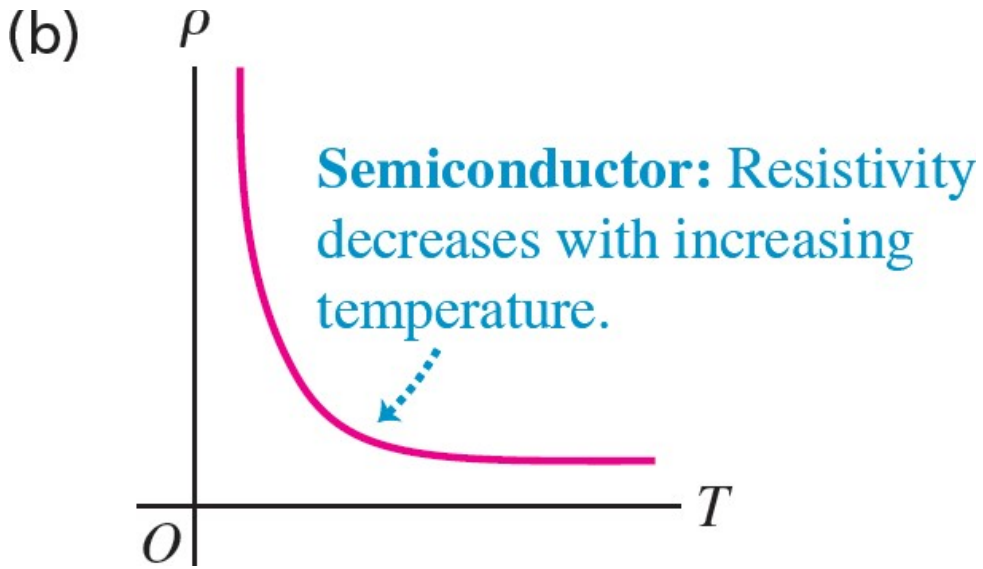
| **Material** | $\alpha[(°C)^{-1}]$ |
|---|---|
| Aluminum | 0.00039 |
| Carbon (graphite) | −0.0005 |
| Copper | 0.00393 |
| Iron | 0.0050 |
| Lead | 0.0043 |
| Silver | 0.0038 |
| Tungsten | 0.0045 |

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# Resistivity and Temperature (2 of 2)

- The resistivity of graphite (a semiconductor) **decreases** with increasing temperature, since at higher temperatures, more electrons “shake loose” from the atoms and become mobile.
- Measuring the resistivity of a small semiconductor crystal is a sensitive measure of temperature; this is the principle of a type of thermometer called a **thermistor**.

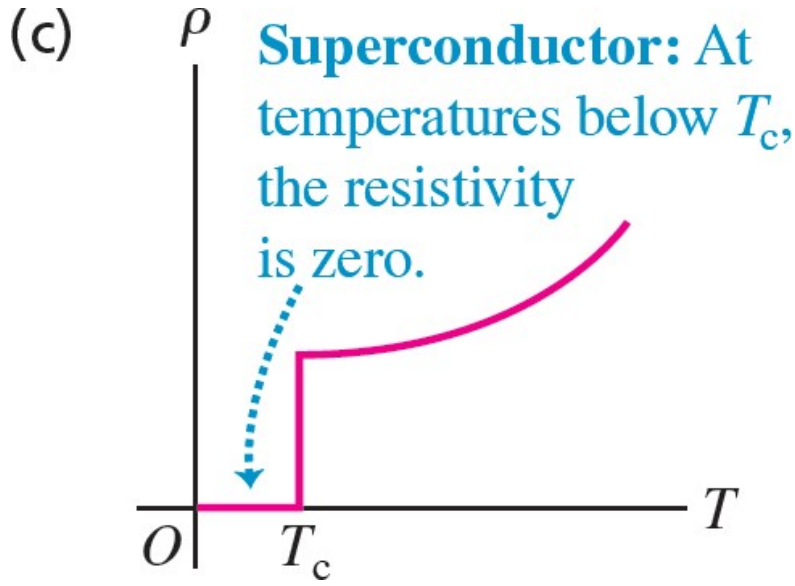



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# Superconductivity

- Some materials show a phenomenon called **superconductivity**.
- As the temperature decreases, the resistivity at first decreases smoothly, like that of any metal. Below a certain critical temperature $T_c$ a phase transition occurs and the resistivity suddenly drops to zero.
- Once a current has been established in a superconducting ring, it continues indefinitely without the presence of any driving field.

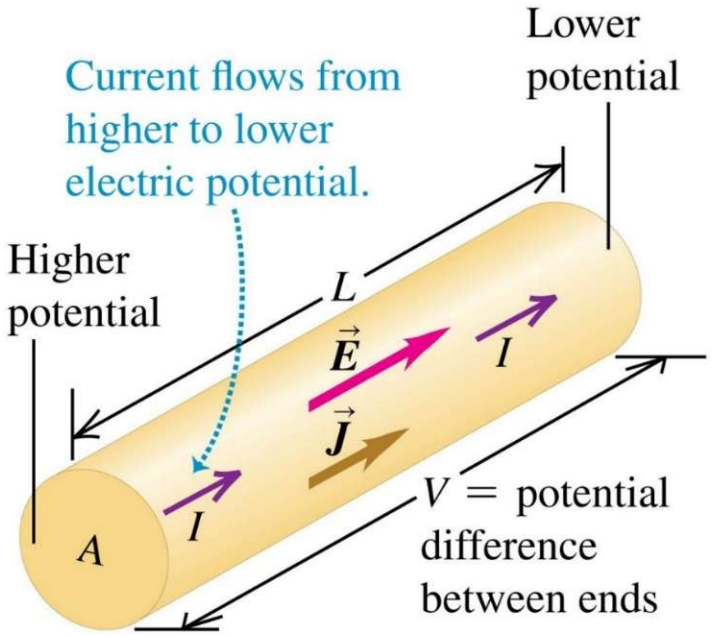



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# Resistance and Ohm's Law

- The **resistance** of a conductor is $R = \rho L/A$.
- The potential across a conductor is given by Ohm's law: $V = IR$.



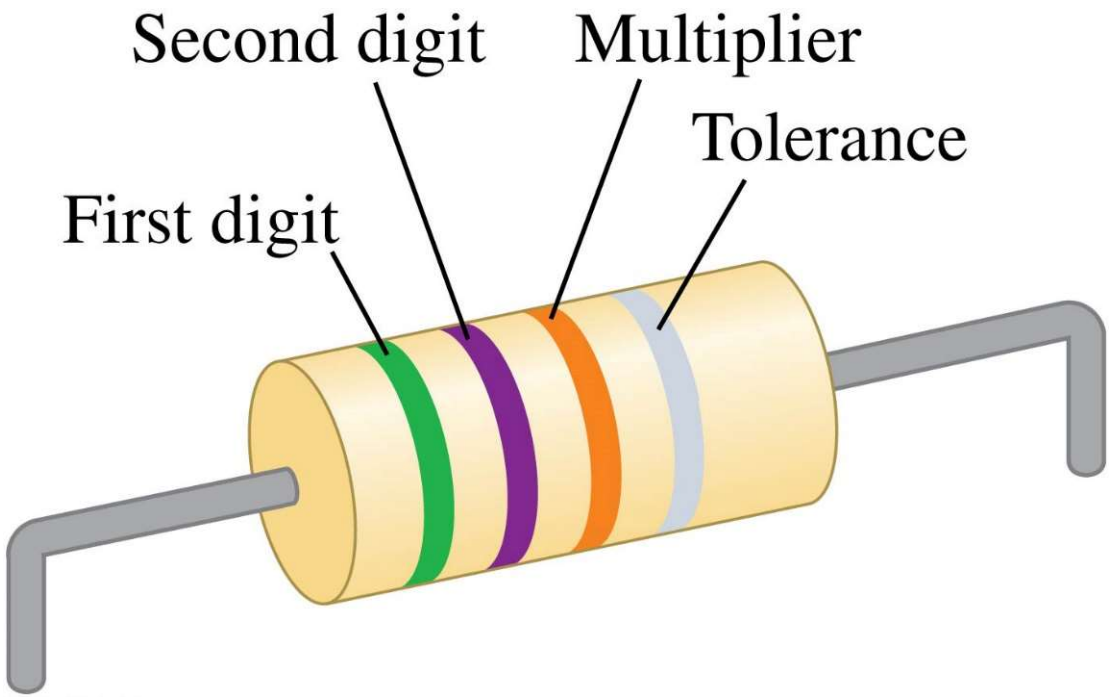

- Video Tutor Demonstration: Resistance in Copper and Nichrome

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# Resistors Are Color-Coded for Easy Identification (1 of 2)

- This resistor has a resistance of 57 kΩ with a tolerance of ±10%.




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# Resistors Are Color-Coded for Easy Identification (2 of 2)

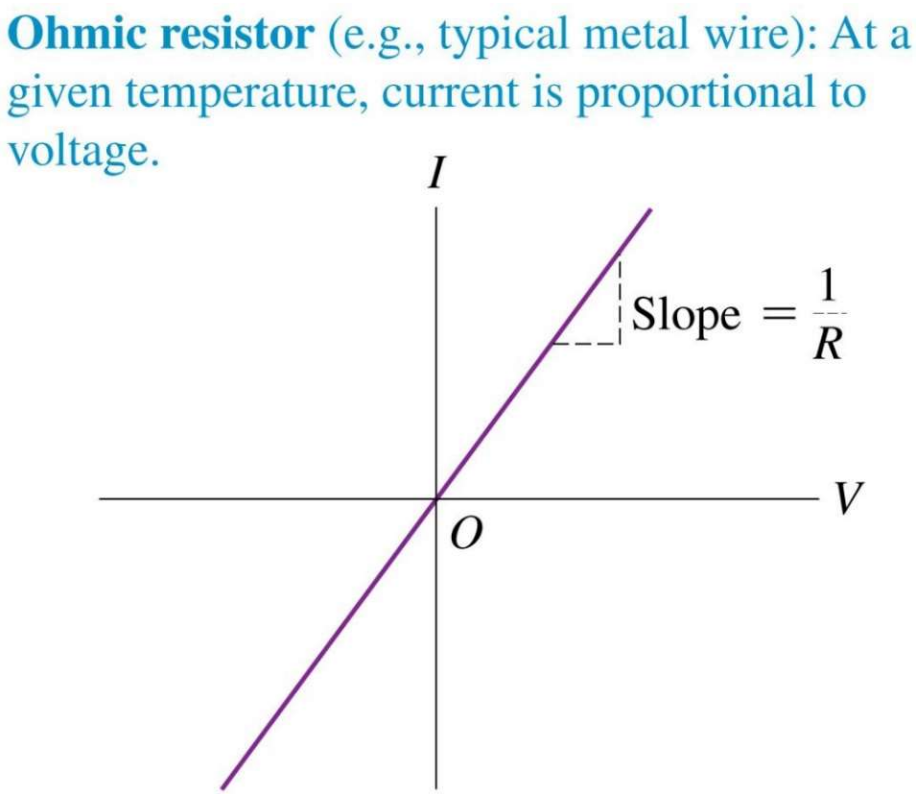
| Color | Value as Digit | Value as Multiplier |
|---|---|---|
| Black | 0 | 1 |
| Brown | 1 | 10 |
| Red | 2 | $10^2$ |
| Orange | 3 | $10^3$ |
| Yellow | 4 | $10^4$ |
| Green | 5 | $10^5$ |
| Blue | 6 | $10^6$ |
| Violet | 7 | $10^7$ |
| Gray | 8 | $10^8$ |
| White | 9 | $10^9$ |

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# Ohmic Resistors

- For a resistor that obeys Ohm's law, a graph of current as a function of potential difference (voltage) is a straight line.



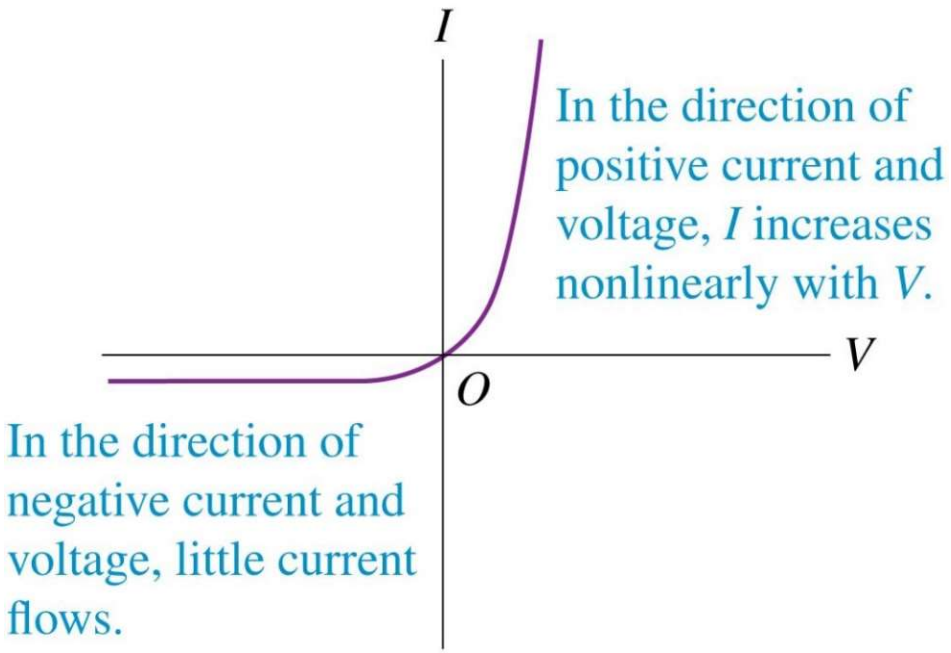

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# Nonohmic Resistors

- In devices that do not obey Ohm's law, the relationship of voltage to current may not be a direct proportion, and it may be different for the two directions of current.

**Semiconductor diode: a nonohmic resistor**




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# Electromotive Force and Circuits (1 of 2)

- Just as a water fountain requires a pump, an electric circuit requires a source of **electromotive force** to sustain a steady current.



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# Electromotive Force and Circuits (2 of 2)

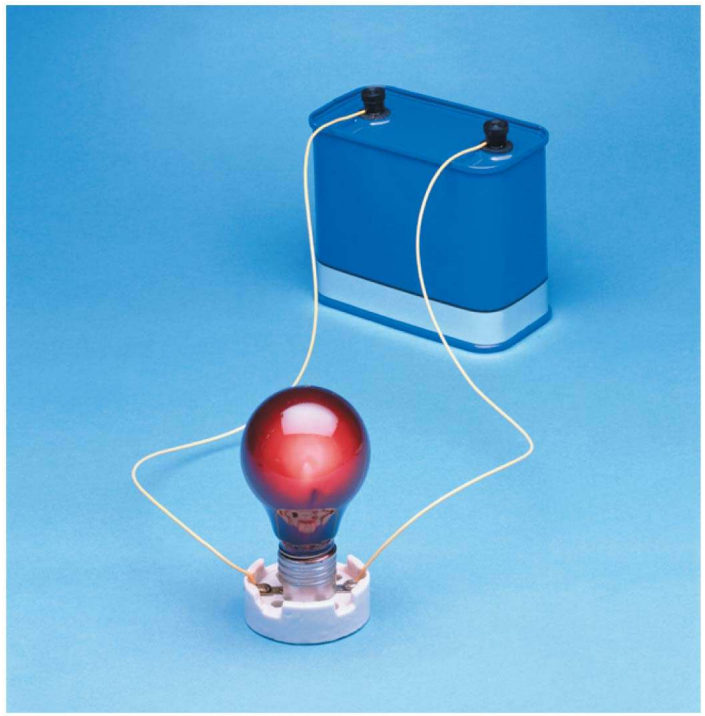
- The influence that makes current flow from lower to higher potential is called **electromotive force** (abbreviated **emf** and pronounced “ee-em-eff”), and a circuit device that provides emf is called a **source of emf**.
- Note that “electromotive force” is a poor term because emf is not a force but an energy-per-unit-charge quantity, like potential.
- The SI unit of emf is the same as that for potential, the volt (1 V = 1 J/C).
- A typical flashlight battery has an emf of 1.5 V; this means that the battery does 1.5 J of work on every coulomb of charge that passes through it.
- We’ll use the symbol $\mathcal{E}$ (a script capital E) for emf.

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# Internal Resistance

- Real sources of emf actually contain some **internal resistance** *r*.
- The **terminal voltage** of the 12-V battery shown at the right is less than 12 V when it is connected to the light bulb.



**Terminal voltage, source with internal resistance** — emf of source — Current through source — Internal resistance of source

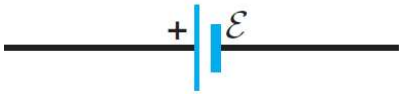
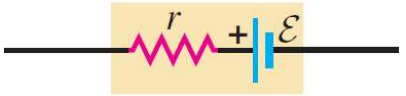
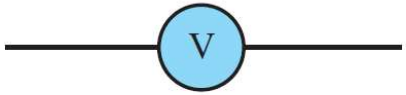
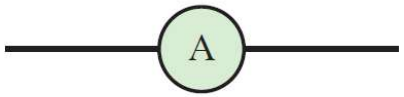
$$V_{ab} = \mathcal{E} - Ir$$

- Video Tutor Solution: Example 25.5

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# Table 25.4 Symbols for Circuit Diagrams

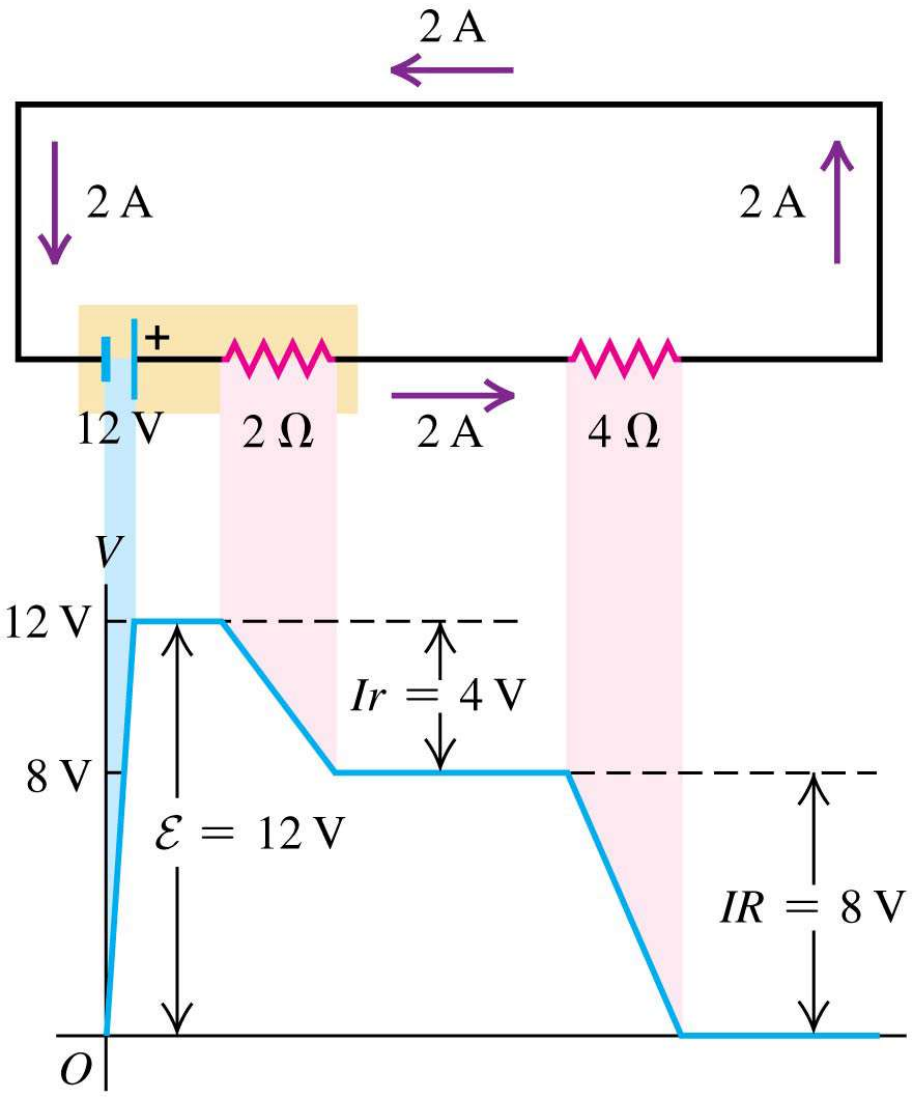
| | |
|---|---|
|  | Conductor with negligible resistance |
|  | Resistor |
|  | Source of emf (longer vertical line always represents the positive terminal, usually the terminal with higher potential) |
|  | Source of emf with internal resistance $r$ ($r$ can be placed on either side) |
| or | |
|  | |
|  | Voltmeter (measures potential difference between its terminals) |
|  | Ammeter (measures current through it) |

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# Potential Changes

- The figure shows how the potential varies as we go around a complete circuit.
- The potential rises when the current goes through a battery, and drops when it goes through a resistor.
- Going all the way around the loop brings the potential back to where it started.

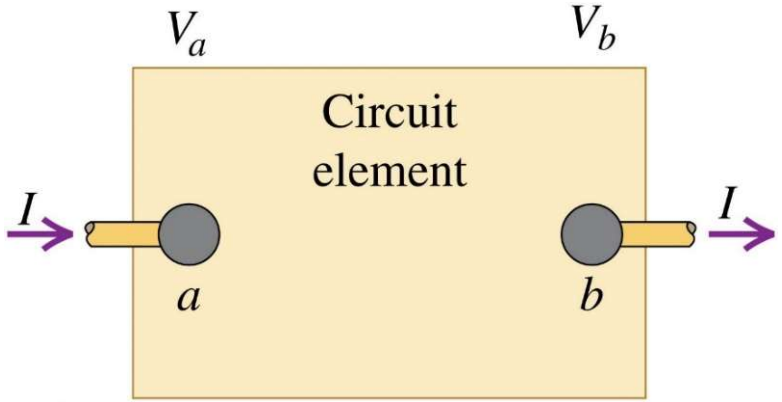



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# Energy and Power in Electric Circuits

- The box represents a circuit element with potential difference $V_{ab} = V_a - V_b$ between its terminals and current $I$ passing through it in the direction from $a$ toward $b$.
- If the potential at $a$ is lower than at $b$, then there is a net transfer of energy out of the circuit element.
- The time rate of energy transfer is power, denoted by $P$, so we write:




**Power delivered to or extracted from a circuit element** $P = V_{ab} I$ (Voltage across circuit element; Current in circuit element)

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# Power

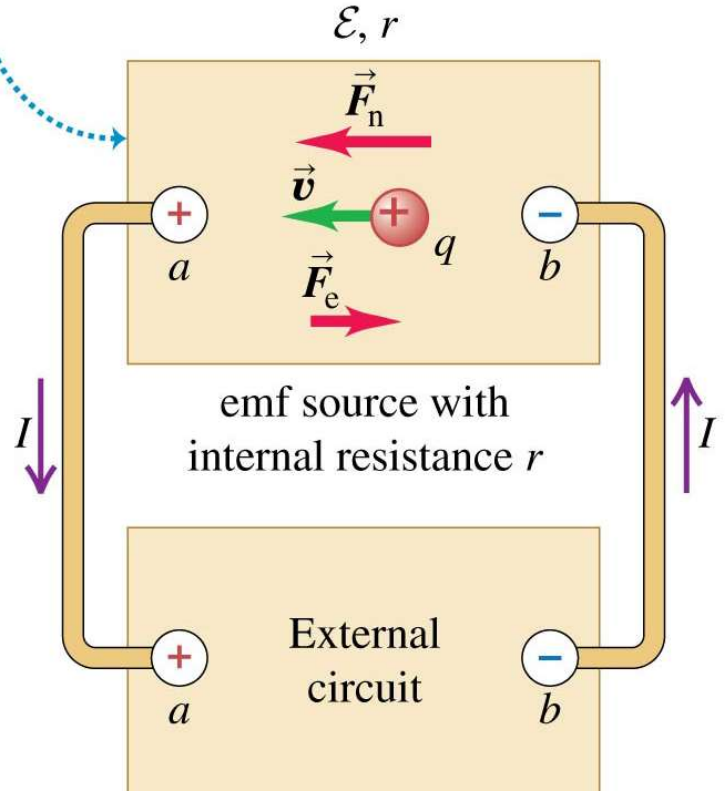
- The upper rectangle represents a source with emf $\epsilon$ and internal resistance $r$, connected by ideal wires to an external circuit represented by the lower box.
- Point $a$ is at higher potential than point $b$, so $V_a > V_b$ and $V_{ab}$ is positive.

$$P = V_{ab} I$$

$$P = \mathcal{E} I - I^2 r$$

- Video Tutor Solution: Example 25.8

- The emf source converts nonelectrical to electrical energy at a rate $\mathcal{E}I$.
- Its internal resistance *dissipates* energy at a rate $I^2 r$.
- The difference $\mathcal{E}I - I^2 r$ is its power output.

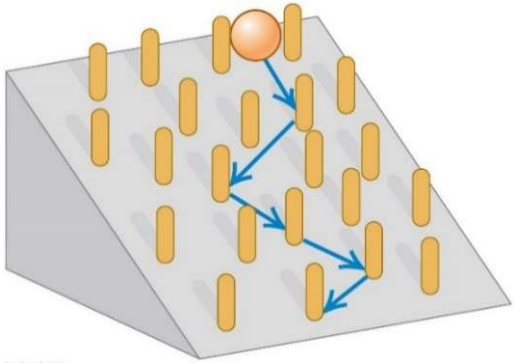


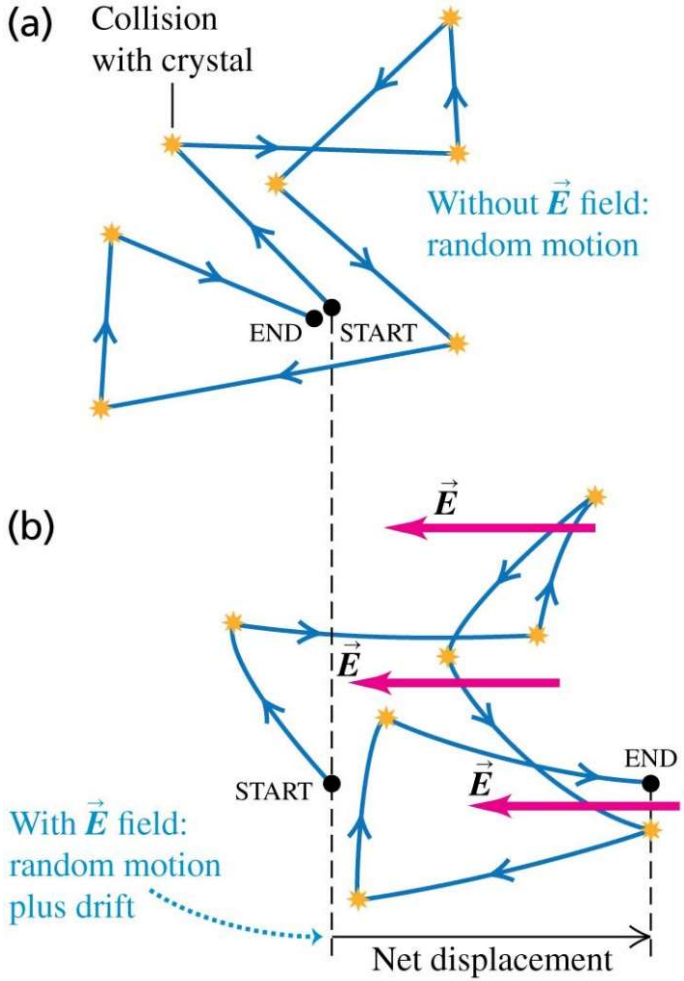

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# Metallic Conduction

- Electrons in a conductor are free to move through the crystal, colliding at intervals with the stationary positive ions.
- The motion of the electrons is analogous to the motion of a ball rolling down an inclined plane and bouncing off pegs in its path.






- Video Tutor Solution: Example 25.11

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